

Gravity Preloading for Maintaining Hair Shape Using the Simulator as a Closed-box Function

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Figure 1: Left: Original shape of groom. Middle: Settling result using preloaded rest shape. Right: Settling result without preloaded rest shape.

ABSTRACT

In animation, hair styles can often be modeled without the consideration of physics. One of the side effects of this workflow is that external forces such as gravity will deform the groom from the designed shape when simulated. We present a simple optimization algorithm that preloads the rest shape to compensate for the external forces to maintain the groom shape during simulation. The algorithm provides artistic control over how much force is compensated for at each vertex.

KEYWORDS

Simulation, groom, hair

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1 INTRODUCTION

Maintaining the groomed shape of hair during simulation is a challenge in production environments. If the modeled groom shape is used as the rest configuration in the simulation, the hair will not maintain the original form due to the influence of external forces, such as gravity. This problem can be solved using optimization techniques that find the rest configuration that will settle to the

modeled groom shape when external forces are applied. However, previous works [Chen et al. 2014; Iben et al. 2019; Twigg and Kacic-Alesic 2011] require knowledge of the material model to solve for the rest configuration.

We propose an algorithm that combines progressive optimization with a novel deformation-gradient-based search direction correction technique. This approach treats the hair simulation as a closed-box function, which makes it relatively easy to implement and adapt to different material models.

2 APPROACH

We first introduce a mathematical formulation of the problem as an optimization and then present our technique for estimating a search direction. We use $\mathbf{x}_{goal}$ to denote the goal shape, or groom shape. This is the shape that we wish to hold under the external forces. We denote the forward simulation function as S . It takes in an external force $\mathbf{f}_{ext}$ and rest shape $\mathbf{X}$, and returns the deformed shape $\mathbf{x}$ by performing a quasi-static simulation:

$$S(\mathbf{X}, \mathbf{f}_{ext}) = \mathbf{x} \quad (1)$$

Our goal is to solve for $\mathbf{X}$ in the following equation:

$$S(\mathbf{X}, \mathbf{f}_{ext}) = \mathbf{x}_{goal} \quad (2)$$

If the current rest shape is $\mathbf{X}^*$, and it settles to $\mathbf{x}^* = S(\mathbf{X}^*, \mathbf{f}_{ext})$, to solve for Equation 2, we linearize the simulation function:

$$S(\mathbf{X}, \mathbf{f}_{ext}) \approx S(\mathbf{X}^*, \mathbf{f}_{ext}) + \frac{\partial S}{\partial \mathbf{X}} \delta \mathbf{X} = \mathbf{x}_{goal} \quad (3)$$

Which can be reordered as follows:

$$\frac{\partial S}{\partial \mathbf{X}} \delta \mathbf{X} = \mathbf{x}_{goal} - S(\mathbf{X}^*, \mathbf{f}_{ext}) \quad (4)$$

$$\frac{\partial \mathbf{x}}{\partial \mathbf{X}} \delta \mathbf{X} = \mathbf{x}_{goal} - \mathbf{x}^* \quad (5)$$

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2.1 Deformation Gradient-Based Derivative Estimation

We want to solve for δX in Equation 5. As formulated, $\frac{\partial \mathbf{x}}{\partial \mathbf{X}}$ is dependent on the material model. We seek a general solution that can be applied to any given Lagrangian simulator. To this end, we present an approximation to the derivative that is independent of the material model.

In continuum mechanics, the deformation gradient is defined as

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} \quad (6)$$

Note that $\mathbf{x}$ and $\mathbf{X}$ in Equation 2 are discretized degrees of freedom, not continuum fields, however the two expressions are symbolically similar. This observation inspired us to use the deformation gradient $\mathbf{F}$ as the estimation for $\frac{\partial \mathbf{x}}{\partial \mathbf{X}}$ in Equation 5.

2.2 Implementation Details

To ensure that the linearization is valid, the deformation needs to be small at each step. Instead of trying to compensate for all of the external forces at once, the external force is progressively increased over n steps, empirically chosen to be 100 in our implementation (Algorithm 1).

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input:  $\mathbf{x}_{goal}$ , external force  $\mathbf{f}_{ext}$ , step size  $\alpha$ , threshold  $h$ 
for  $i = 1$  to  $n$  do
     $\mathbf{f} = \mathbf{f}_{ext} * i/n$ ;
    while  $|\mathbf{S}(\mathbf{X}, \mathbf{f}) - \mathbf{x}_{goal}| > h$  do
         $d\mathbf{x} = \mathbf{x}_{goal} - \mathbf{S}(\mathbf{X}, \mathbf{f})$ ;
         $\mathbf{X} += \alpha * \mathbf{F}^{-1} d\mathbf{x}$ ;
    end
end

```

Algorithm 1: Preloading Algorithm

There is no guarantee of convergence for the algorithm. In practice, the algorithm converges in most cases under a moderate step size (0.1). We have implemented a fail safe method to handle the cases when the algorithm does not converge. In most hair simulations, each hair curve is fixed at the root, and free at the tip. As a result, the points near the root hold the weight of the whole curve while the points near the tip hold little weight. The points near the root will therefore require a larger amount of deformation to balance the force they bear. When the algorithm fails to converge, it is usually the root points that fail first. Based on this observation, when the inner loop of Algorithm 1 fails, the fail safe fixes the points starting from the roots one by one as Dirichlet conditions that match the goal shape until the inner loop converges. Those points will remain fixed in the outer loop.

2.3 Artistic Control

When the preloaded strain completely compensates the gravity, the hairs are at a close force equilibrium, and the resulting motion can be very energetic and bouncy. Therefore in practice, we have found it works best to utilize some combination of preloading and collisions between layers of hair to maintain the groom under motion.

Artists can control how much gravity is compensated at each vertex through a map. The map is required be monotonically increasing from root to tip. If the map value at a vertex is less than the current step i in the outer loop, the vertex is fixed as a Dirichlet condition in the inner loop.



Figure 2: Selected frames of two simulations, without (left) and with (right) preloading

3 RESULTS

Figure 2 illustrates how preloading the rest shape helps keep the hair from falling in front of the character's face, and prevents loss of volume, maintaining the silhouette. The example shown has 10034 vertices, and the preloading process ran for 5.2 seconds on a 24 core machine. In practice, preloading is run once per hair rig build, and can optionally be re-run based on the particular needs of a shot.

4 CONCLUSIONS AND FUTURE WORK

We have presented an optimization algorithm that utilizes forward simulation as a closed-box function and demonstrated the approach on production assets. The algorithm can be slower than comparable methods that utilize the knowledge of the material model, but our method has the advantage that it is agnostic to the material model, which means it can be directly reused for any new material with minimal modifications. This approach was first used on selected characters in Disney's "Raya and the Last Dragon", and is now incorporated into most hair simulation rigs. For future work, we plan to test the algorithm with more material models (e.g. cloth).

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